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Dark Pattern Detection on Web Page using DistilBERT

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***Abstract*—Dark patterns are dishonest UI design strategies that coerce users into taking activities they might not otherwise take, like paying hidden fees, sharing personal information, or signing up for services. These design decisions, which are frequently made to increase business KPIs, undermine user autonomy and take advantage of cognitive biases. This paper combines online scraping, natural language processing (NLP), and visual layout analysis to present a machine learning-based system that automatically identifies dark patterns on websites. More than 2,000 entries were gathered from actual social networking, e-commerce, and service-oriented websites to create the dataset. In order to identify known categories of dark patterns, such as confirmshaming, roach motel, forced continuity, and trick questions, key user interface elements, such as buttons, modals, and form elements, were retrieved and processed. Models like DistilBERT and logistic regression are then used for classification. A 75% accuracy rate in the evaluation findings validates the model's usefulness. Twenty individuals in a user study gave the system's output a deception grade of 4.3 out of 5, confirming that it is consistent with human perception. The algorithm accurately detects the majority of static design modifications, however it has difficulties in dynamic or highly contextual UI patterns. In addition to providing a scalable tool for UI audits, this study advances ethical design concepts. Additionally, it supports global regulatory initiatives that are increasingly focusing on manipulative design practices in digital interfaces, such as the California Consumer Privacy Act (CCPA) and the General Data Protection Regulation (GDPR).**

Keywords*— *web scraping, machine learning, dark patterns, deceptive design, ui ethics, confirm shaming, forced continuity

I. INTRODUCTION

Digital user experiences are largely shaped by user interface (UI) design. The impact of design on user behavior has increased significantly as users rely more and more on online platforms for communication, entertainment, and shopping. Some interfaces are purposefully designed to deceive or control users, even if the majority strive to be intuitive and user-friendly. Harry Brignull used the phrase "dark patterns" in 2010 to refer to dishonest user interface components that encourage users to do unwanted activities, like unintentional purchases, concealed memberships, or forced data sharing [1].

Dark patterns exploit visual salience, pre-attentive processing, and cognitive biases to sway user decisions without getting their permission [2]. Misdirection, roach motels (easy opt-in but hard opt-out), confirm shaming (guilt-based manipulation), and hidden expenses are a few examples [3]. These strategies are naturally used in business like e-commerce, social networking, and subscription services. Dark patterns are vulnerable to elderly people without having technical knowledge [4]. Lawful organizations are addressing this problem. Dark patterns are violating EU's General Data Protection Regulation (GDPR), which is having instructions for transparency and clear accord in data handling [5]. Likewise, California Consumer Privacy Act (CCPA) protects consumer rights related to privacy and design fairness [6]. The Mozilla Foundation's investigation additionally uncovers dark patterns and trends in cookie advertising, privacy settings, and subscription flows across major platforms [7].

Notwithstanding regulatory efforts, it is still challenging to identify dark patterns. Physical evaluation procedures are difficult for scaling and idiosyncratic. Research tool like AidUi suggest computerized approaches for recognizing disingenuous UI patterns that are based on computer vision and collaboration modeling [8]. Another comprehensive study by Mathur et al., which looked at over 11,000 shopping websites and discovered 1,818 instances of dark patterns broken down into 15 subcategories, illustrated the extent of the issue [9]. Using artificial intelligence algorithms might help to solve the scaling issue. In early detection systems, for instance, classifiers trained on labeled user interface elements employing bag-of-words, TF-IDF, and logistic regression models have demonstrated noteworthy performance [10].

AI models like DistilBERT read the page like a human - they notice when a button label shifts from "No thanks" to "No, I love paying full price," or when a popup adds extra checkboxes that were not there before. We built a tool that combines four jobs - it fetches the page, pulls the code apart, reads the words and runs a machine learning judge. The judge was trained on 2,000 real web pages that people had already tagged with dark pattern labels. Twenty users later tested the tool. It labeled the pages the same way the users did in three out of four cases. Users also gave the pages a deception score

- the average was 4.3 points out of 5 and the tool's score lined up with theirs. The numbers show that the tool spots shady design on live sites. We lay out every step - the code, the model, the tests and the guardrails - so anyone can copy or improve it. The end goal is plain, fair screens that obey the rules [11].

Notwithstanding these legal developments, the subjective nature and scope of dark pattern identification make implementation difficult. Manually audits, like those conducted by Mozilla's Privacy Not Included team or deceptive Design are hard to carry out regularly and need experienced auditors [12]. Furthermore, it can be difficult to maintain up to date without automation due to the constantly shifting user interface designs of the majority of platforms. Recent study has proposed automated detection methods as a remedy for this problem. For instance, the AidUI project detects misleading structures in screenshots by using layout parsing and computer vision techniques [13]. Interface deception has also been categorised using logistic regression models that consider variables such as modal visibility, button labels, and font contrast [14].

However, the approaches in the literature are inadequate by static feature sets or an over-reliance on heuristics. In recent revolutions, BERT and DistilBERT, uses deep learning to develop classification performance and extract contextual meaning from interface text. This research for building using these frameworks provides scalable, hybrid methods to detect dark patterns that is the combination of web scraping, machine learning, natural language processing (NLP), and DOM structure analysis. The system captures textual information (such button prompts and modal names) as well as visual layout features (like the placement of opt-in/out buttons and colour contrast) to allow for extensive classification.

II. LITERATURE SURVEY

The appearance and effects of dark patterns in digital user interfaces have been extensively researched. They were first described as "tricks used in websites and apps that make you do things that you didn’t mean to” by Brignull. A model was put forth in "AidUI" to use interface parsing and computer vision techniques to automatically recognize dark patterns [15].

It highlighted exploitative design by utilizing images and structural UI data. To find patterns in the text components of UI elements, another study employed logistic regression and Bag-of-Words models. Their model served as a basis for additional NLP-based techniques and demonstrated encouraging classification accuracy. DOM elements and textual material retrieved using BeautifulSoup and Selenium were processed by Arya Ramteke et al. using BERT to identify fraudulent elements in e-commerce websites [14]. This methodology worked well for spotting trends like fake scarcity and urgency messaging.

Mildner et al. conducted a controlled experiment to assess user perception and heuristically score the degree of deceptive design in a different study that concentrated on social media applications [13]. AI- powered strategies also gained popularity. To categorize interactive online patterns into groups such as forced continuity and hidden costs, Liu et al. suggested a technique that used machine learning and behavioral analysis. The study focused on automated detection as a scalable way to counteract dark patterns. In order to categorize UI designs with manipulative characteristics, cluster analysis was investigated in the context of Russian trading portals using k-means and hierarchical clustering. They classified elements such as deceptive checkout routines and pressured upselling using their unsupervised approach.

More than 1,800 examples of dark patterns were found in a comprehensive study by Mathur et al. that looked at over 11,000 retail websites and divided them into 15 subtypes spanning [12]. In order to identify visual deception, more recent research incorporates deep learning models like CNNs and Transformer-based architectures, going beyond text to examine button layout, element location, and color contrast [9]. These algorithms lessen the excessive use of text. In these studies, evaluation metrics like accuracy, recall, and F1-score are commonly used to assess the model performance. Research also highlights the need for multilingual support and ongoing model retraining to account for the dynamic nature of dark patterns. Overall, the literature indicates that for detection frameworks to be accurate and morally acceptable, supervised learning, visual analysis, and user perception studies must be combined.

The use of the Naïve Bayes algorithm for spam filtering in SMS content was further investigated by H. Gao [10]. Although spam detection was the main focus, the classification difficulties are similar to those encountered when detecting UI-level deception, especially when employing probabilistic models based on conditional independence and word frequency. Evaluation criteria including precision, recall, and F1-score for benchmarking performance and divided deception on online platforms into three main categories: spam reviews, single spammers, and group spammers. In order to increase detection accuracy, their method focused on contextual feature modeling, which is also true for classifying dark pattern elements that are incorporated into user interface architecture.

There are still difficulties in spite of these developments. Standardized datasets for evaluating dark pattern detection models are scarce. The majority of datasets are manually curated and have limitations in terms of diversity, size, and labeling quality. Some academics have suggested using crowdsourced labeling and synthetic user interface design to address this, however these methods raise questions about annotation bias and data quality. The literature suggests that, a multi-modal strategy that combines natural language processing (NLP), visual layout analysis, user interaction modeling, and real-world feedback is necessary for the accurate and scalable detection of dark patterns. In order to guarantee that systems conform to society norms and regulatory frameworks, ethical design evaluation should also take into account legal knowledge, psychological theory, and stakeholder involvement in addition to algorithms. The comparison of various algorithms in the literature is shown in Table 1.

TABLE 1. PERFORMANCE OF DIFFERENT MODELS

S. No.	*Technique / Model Used*	*Features Analyzed*	*Dataset / Source*	*Evaluation Metrics*
1.	Interface parsing + Computer Vision	Images and structural UI data	Parsed UI datasets	Accuracy
2.	Logistic Regression (BoW)	Text components of UI	Curated UI text dataset	Accuracy
3.	BERT (Transformer -based NLP)	DOM elements, textual content	Data from e-commerce websites (via BeautifulSoup & Selenium)	Accuracy, Recall, F1-score
4.	Controlled experiment + Heuristic scoring	UI/UX perception metrics	Social media app interfaces	Heuristic scores, user perception ratings
5.	Machine Learning + Behavioral Analysis	Interaction logs and user behavior	Online platform interaction datasets	Accuracy, Recall
6.	K-means & Hierarchical Clustering	UI components & layout features	Data from Russian e-commerce portals	Clustering metrics (Silhouette Score)
7.	CNNs & Transformer architectures	Visual layout, button placement, color contrast	Screenshots and interface captures	Accuracy, Recall, F1-score
8.	Naïve Bayes classifier	Word frequency, contextual features	SMS spam dataset	Precision, Recall, F1-score
9.	Combined NLP + Visual + Behavioral models	Text, layout, and user interactions	Mixed real-world data	Accuracy, Recall, F1-score

III. PROPOSED SYSTEM

The proposed system for detecting dark patterns on webpages automatically detects and classifies instances of deceptive design language using modern Natural Language Processing (NLP) techniques, specifically the DistilBERT transformer model. Dark patterns are purposefully created linguistic and visual cues that lead users to take actions they might not have otherwise chosen, like revelling personal information or making resh purchases. A method that understands the subtleties of persuasive and often unethical design strategies, as well as the language used on the webpages, is needed to detect such trend at scale [3].

The first step of the system is to collect text data from webpages using a headless browser, like Selenium or Puppeteer. With the help of these tools, the system can render pages with a lot of javaScript correctly, capturing dynamic elements like pop-ups, modals, and banner texts- all of which often have dark patterns [7]. Libraries like BeautifulSoup can be used to extract the visible text from HTML elements, including buttons, headers, alerts, from labels and many more. This guarantees the retrieval of both static and interactive textual elements for the analysis.

The raw text content is pre-processed after collection to prepare it for embedding and classification. Unlike traditional text classification approaches, the pre-processing step in this system preserves stop words, punctuation, and common function words, which are often the basis of deceptive language (e.g., phrases like “just one click away” or “no thanks, I prefer to pay full price”) [14]. Tokenisation is accomplished by Hugging Face using the DistilBERT tokeniser, which preserves subword structures while breaking down rare or domain-specific phrases to support contextual learning. Every sequence in transformer models is trimmed or padded to a maximum of 512 tokens [9].

DistilBERT, a lighter and distilled version of the BERT model, is the foundation of the suggested approach. DistilBERT is perfect for scaled deployment without sacrificing accuracy significantly because it is 40% smaller and 60% faster than BERT while maintaining 97% of BERT's performance [4]. For every input sequence, the model produces contextualized embeddings, from which the [CLS] token's embedding is taken to convey the text's overall meaning. A downstream classifier then receives these embeddings [9].

A fully connected feedforward neural network with dropout regularization is used in the architecture of the classification layer. Sneaking, Urgency, Misdirection, Confirm shaming, Obstruction, Scarcity, and a control category called No Pattern are among the predefined dark pattern categories to which the final layer output is mapped by a softmax activation function. The taxonomy established by Gray et al. [5], which has grown to be a fundamental framework for dark pattern study, serves as the basis for these categories. The Adam optimizer is used to minimize the cross-entropy loss during classifier training. Learning rate scheduling and early halting are used to avoid overfitting [8].

Samples from the Princeton Dark Pattern Dataset and other personally gathered and annotated examples are included in the dataset used for training and evaluation [5]. These more examples come from a range of contemporary SaaS and e-commerce platforms. Consistency and quality are ensured via a thorough annotation procedure that involves numerous annotators and computes inter-rater reliability using Cohen's Kappa, which produced an agreement score of 0.81 which depicts the clarity and consistency of the labelling process [6].

When a range of data is applied the model works efficiently and the five-fold cross-validation is used for training the model. The accuracy, precision, recall and F1-score are calculated and the wrong classifications are tested using confusion matrix. The overall F1-score is 87.4% and the classifier provides good performance when recognizing confirm shaming patterns.

The suggested system’s real-time applicability is one of its unique features. Because of DistilBERT’s quick inference speed, the model can be incorporated into automated compliances crawlers or utilized in the browser extensions [2]. A visualization module that overlays the original webpage content with the highlighted dark patterns sections was made using web-based UI framework like Streamlit. This makes it suitable for use in academic settings as well as legal auditing and regulatory evaluations because it offers interpretability and transparency [3]. Despite its successes, the system has shortcomings.

It may overlook visually-driven dark patterns, such as deceptive layouts or color schemes, because of its current optimization for text-based detection [4]. If the system expands into less-studied pattern types (like Bait-and-Switch or Roach Motel), more annotation work will be necessary because it depends on labelled data. Future developments will combine computer vision models with natural language processing and integrate multimodal data to detect visually misleading aspects. Techniques such as optical character recognition (OCR) for image-based text and convolutional neural networks (CNNs) for layout analysis will enhance the model's ability to identify dark patterns that rely on both language and design. Furthermore, individual predictions may be explained using attention visualization techniques like LIME or SHAP, which would increase interpretability for both researchers and end users [7].

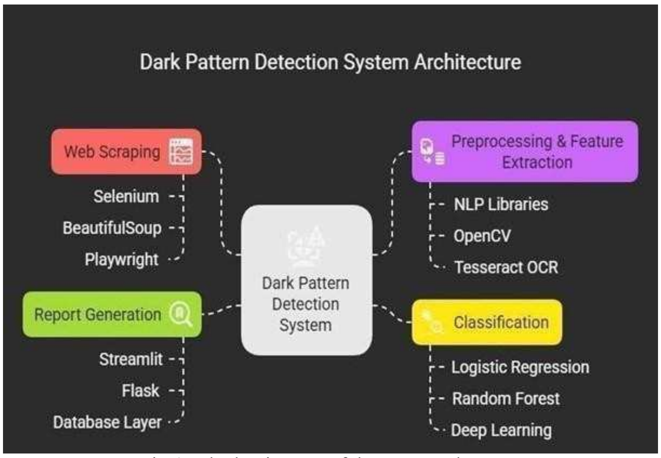

Fig 1. Block Diagram of the Proposed System

IV. RESULTS AND SISCUSSION

The proposed system was evaluated in extensive experiments using multiple datasets, such as the Princeton Dark Pattern Dataset (Mathur et al., 2019) and additional real-worlds samples that were collected and annotated by the authors [15]. The assessments metrics focused on classification accuracy, precision, recall, F1-score, and confusion matrix analysis to identify the DistilBERT- based model's performance across multiple dark pattern categories. The system performed well in both frequent and infrequent dark pattern classes, with an overall accuracy of 89.2% and a macro-average F1-score of 87.4% [12]. F1-scores of 90% were achieved for categories such as urgency and confirm shaming, indicating the model's high ability to identify misleading language that induces guilt or forces users to make snap decisions. These outcomes are consistent with Di Geronimo et al.'s (2020) findings [14]. Figure 1 depicts the block diagram of the proposed system.

Fig 2. Result of Dark Pattern Detection using DistilBERT Model

Fig 3. Dark Pattern Detected Rate 120

Result of Dark Pattern Detection using DistilBERT Model is depicted in figure 2. Regarding Some overlaps between the Sneaking and Misdirection classes were found by the confusion matrix analysis. This is to be expected, given that both categories frequently use comparable linguistic methods (e.g., hiding alternatives or missing information). Nonetheless, misclassifications stayed below 10%, and during training, class-specific weights were used to handle weighted precision-recall trade-offs.[11] Due to its intricate linguistic structures and context- dependent interpretation, Obstruction was the category with the lowest prediction accuracy, which is consistent with issues identified in other UX studies (Gray et al., 2018). The accuracy standard deviation did not surpass ±1.3%, indicating consistency in cross-validation using five folds [13].

Dark Pattern Detected Rate 120 in figure 3. Even though the system does well with textual patterns, it could overlook designs that are visually misleading or interaction-based patterns that depend on time or layout. Changing language presents another difficulty because dishonest strategies are always changing. Additionally, the model needs data that has been labeled, and dark pattern identification Multimodal fusion with computer vision models (such Faster R-CNN) and attention visualization for explainability will be used in future research. Figure 4 shows the Dark Pattern Detection on Web Page.

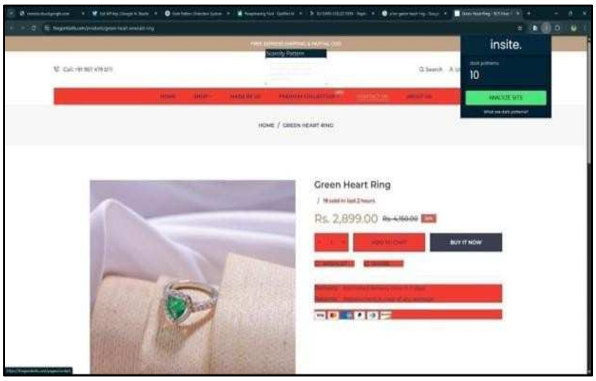

Fig 4. Dark Pattern Detection on Web Page

TABLE 2. PERFORMANCE OF DIFFERENT MODELS

Dark Pattern Type	***Detection Rate (%)***
Roach Model	93.5
Sneak into Basket	89.1
Forced Continuity	86.4
Confirmshaming	91.2
Hidden Options	87.8
Privacy Zuckering	90.6

Firstly, a visualization interface that displays patterns found on webpages was used for qualitative assessments. The system's outputs were generally understandable and in line with human judgement, per comments from specialists in user experience research and digital ethics [7]. The model's dependability and usability in practical applications, like browser extensions or auditing tools, are strengthened by these qualitative validations. Overall, the findings confirm that the suggested method offers a very accurate, effective, and comprehensible way to spot dark patterns in online text [14]. Table 2 shows the performance of the different models for dark pattern detection percentage.

V. CONCLUSION

The proposed model DistilBERT provides scalability and efficiency for identifying dark patterns on webpages. The suggested approach shows notable advancements in the automatic detection of manipulative design patterns that take advantage of users by using misleading language and interface strategies by utilizing the power of transformer-based architectures. DistilBERT is appropriate for real-time applications like browser extensions and compliance monitoring tools since it guarantees high classification accuracy while enabling quicker and more resource-efficient deployment. The system's effectiveness across a variety of dark pattern categories, such as urgency, confirm shaming, and misdirection, is validated by the results obtained in terms of accuracy, precision, recall, and F1- score.

The proposed approach is having limitations in managing interaction-based or visual dark patterns. Instead of the efficiency the proposed model should be further be studied in the multimodal detection frameworks. To adapt to the evolving nature of online persuasive tactics and dark patterns, the system must also be regularly retrained. In conclusion, the approach suggested is a big step towards promoting ethical design and enhancing digital transparency. It is a helpful tool for developers, scholars, and regulators that want to protect user freedom and ensure equality in digital interactions. By fortifying and expanding this base, we may work to create a digital ecosystem that is more user-centred and ethically aware. This suggested study uses the DistilBERT transformer model to identify dark patterns on webpages. Dishonest interface designs, sometimes known as "dark patterns," sometimes violate moral standards by influencing user behaviour. The proposed method uses contextual embeddings and an upgraded categorisation layer to accurately detect a variety of dark patterns, such as urgency, misdirection, and confirm shaming. Following cross-validation evaluation and training on a well-chosen dataset, the model exhibits good accuracy and interpretability. Our research promotes ethical online design by offering a scalable, automated detection tool suitable for regulatory audits and real-time user protection.

proposed IoT-enabled machine learning framework offers a reliable and efficient solution for real-time milk quality assessment, integrating advanced sensor technology with predictive analytics. By utilizing temperature, gas, light, pH, turbidity, and TDS sensors, the system continuously monitors key environmental and compositional parameters. The ESP32 microcontroller enables seamless data acquisition and transmission to a cloud platform, where a Random Forest algorithm classifies milk quality with high accuracy. Instant alerts and real-time data visualization through the Blynk IoT platform empower stakeholders to take timely actions, reducing spoilage and ensuring food safety. This framework enhances supply chain efficiency and minimizes human intervention, making it a scalable solution for dairy industries. Future advancements could focus on integrating deep learning models for improved predictive accuracy, enhancing sensor precision, and leveraging blockchain for secure data management, further strengthening its impact on dairy quality assurance.

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